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American Association of Clinical Endocrinologists
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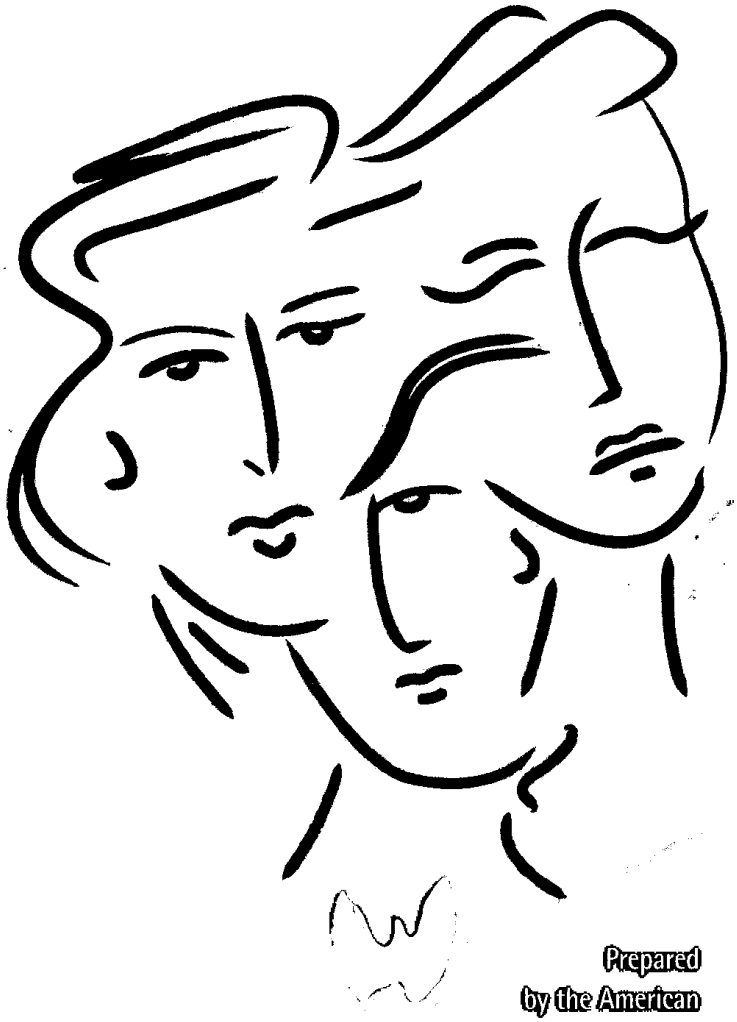
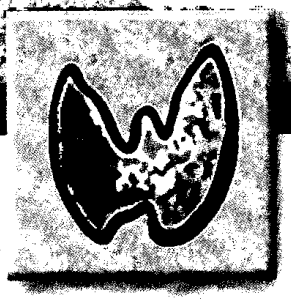
Information for Patients

Hypothyroidism



Prepared by the American Association of Clinical Endocrinologists (AACE), a not-for-profit national organization of specially qualified board-certified endocrinologists whose aims include maintaining the high standards of its physician members and providing high-quality care to patients with endocrine disorders.



What is hypothyroidism?

Hypothyroidism occurs when the thyroid gland produces less than the normal amount of thyroid hormones. The result is the "slowing down" of many bodily functions. Although hypothyroidism may be temporary, it usually is a permanent condition. Some studies have shown that as many as 10% of women and 3% of men have hypothyroidism. At the beginning of this disease, the symptoms are mild and may not be detected. In time, however, the symptoms get more severe, produce discomfort, and eventually may become disabling.

How is hypothyroidism treated?

Hypothyroidism is treated with a single daily dose of levothyroxine, given as a tablet. Various forms are available. An experienced physician can prescribe the correct form and dosage to return the thyroid balance to normal. Levothyroxine replacement therapy is usually started at a low dose and gradually increased. This allows the body to become accustomed to normal thyroid hormone levels. Hypothyroidism is rarely short-lived. Most cases are permanent and often progressive. Therefore, lifelong usage, with changes in dose requirements from time to time, is common. Periodic monitoring of TSH levels and clinical status is necessary to ensure that the proper dose is given. It is extremely important that women planning to become pregnant are kept perfectly well adjusted, since hypothyroidism can affect the development of the baby. During pregnancy, thyroid hormone replacement requirements often change, so more frequent monitoring is necessary. Various medications and supplements (particularly iron) may affect the absorption of thyroid hormone; therefore, the levels may need more frequent monitoring during illness or change in medication.

Inadequate replacement of thyroxine will result in continued symptoms of hypothyroidism and increased serum cholesterol levels. This condition increases the risk for atherosclerosis and coronary artery disease. Excessive replacement of thyroxine can result in symptoms of hyperthyroidism, excessive strain on the cardiovascular system, and possibly an increased risk of developing osteoporosis.

Appropriate management of hypothyroidism requires continued care by an experienced physician.

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How is hypothyroidism diagnosed?

Characteristic symptoms and physical signs, which can be detected by a physician, can signal hypothyroidism. In addition, tests can be used to confirm the diagnosis and to determine the cause of hypothyroidism.

TSH (thyroid-stimulating hormone or thyrotropin) test

An increased TSH level in the blood is the most accurate indicator of hypothyroidism. Production of this pituitary hormone is increased when the thyroid gland even slightly underproduces thyroid hormone.

Other tests

- T_4 (thyroxine) and T_3 (triiodothyronine)—the active thyroid hormones in the blood
- Thyroid autoantibodies — indicates the likelihood of autoimmune thyroiditis being the cause of hypothyroidism

Usually, a general physician can make the diagnosis of hypothyroidism, but assistance is often needed from an endocrinologist, a physician who is a specialist in thyroid diseases.

What are the symptoms of hypothyroidism?

In its earliest stage, hypothyroidism may cause few symptoms other than mild enlargement of the thyroid gland and an increased blood cholesterol level. As thyroid hormone production decreases and the body's metabolism slows, a variety of symptoms may result:

- Fatigue and exhaustion
- Drowsiness
- Forgetfulness
- Difficulty with learning
- Dry, brittle hair and nails
- Dry, itchy skin
- Puffy face
- Constipation
- Sore muscles
- Weight gain
- Heavy menstrual flow
- Increased frequency of miscarriages
- Increased sensitivity to many medications

What causes hypothyroidism?

Autoimmune thyroiditis
(Hashimoto's thyroiditis—separate brochure available)

The body's immune system may produce a reaction in the thyroid gland that results in hypothyroidism and, most often, a goiter (enlargement of the thyroid). Other autoimmune diseases may be associated with this disorder, and additional family members may also be affected.

Radioactive iodine treatment

Hypothyroidism may occur shortly after or many years after the use of radioactive iodine as treatment for hyperthyroidism.

Spontaneous onset

Underproduction of thyroid hormones (hypothyroidism) can occur for no apparent reason.

Thyroid operation

Hypothyroidism may be related to surgery on the thyroid gland, especially if more than one lobe of the thyroid has been removed.

Medications

Lithium, high doses of iodine, amiodarone, and certain resins can cause hypothyroidism.

Subacute thyroiditis

A viral infection of the thyroid, which is usually painful, may cause hypothyroidism after a period of overactivity (hyperthyroidism). Although the hypothyroidism is usually temporary, in a small percentage of cases, it may become permanent.

Postpartum thyroiditis

Shortly after pregnancy, the thyroid may go through a period of hyperthyroidism, followed by hypothyroidism, and then most often returns to normal function.

Congenital condition

An infant may be born with an inadequate amount of thyroid tissue or an enzyme defect that does not allow normal thyroid hormone production. If this condition is not treated promptly, physical stunting and/or mental damage (cretinism) may develop.

Pituitary hypothyroidism

Any destructive disease of the pituitary gland may cause damage to the cells that stimulate the thyroid to produce normal amounts of thyroid hormone. This is a very rare cause of hypothyroidism.